

FutureXcel (Week # 4)

Data Science

Official Task Instructions:

Data Science Interns

Branding rule: Footer on every report: “Analytics by [Name]”.

WEEK 4 — Feature Engineering & Transformation Suite

What is it? Create engineered features to improve analysis/model performance.

What it should contain: 8–10 new features (e.g., ratios, lag features, time-of-day flags, categorical buckets), encoding, scaling, transformations, documentation.

Task description: Generate meaningful features, show distribution changes, and explain the rationale for each feature. Include sample code.

Required work:

- Notebook with transformations & final dataset
- Explanation table: feature name → formula → expected impact
- PDF with before/after model-ready dataset sample

Common problems & fixes: Generating leak features → ensure no future info used; too many sparse features → aggregate categories.

Recommended tools: pandas, sklearn (preprocessing).

